CASE STUDY 53

FOOD WASTE MONITORING SYSTEM

Python Program:

```
print("===== FOOD WASTE MONITORING SYSTEM =====")


days = int(input("Enter the number of days to monitor: "))


if days <= 0:
    print("Error: Number of days must be positive.")


else:
    waste_records = []


    for day in range(1, days + 1):
        waste = float(
            input("Enter food waste in kg for Day {}: ".format(day))
        )


        if waste < 0:
            print("Invalid input: Waste cannot be negative.")
            break


        waste_records.append(waste)
```

```
if len(waste_records) == days:
    total_waste = sum(waste_records)
    average_waste = total_waste / days
    highest_waste = max(waste_records)
    lowest_waste = min(waste_records)

    print("\n===== FOOD WASTE REPORT =====")
    print("Total food waste: {:.2f} kg".format(total_waste))
    print("Average daily waste: {:.2f} kg".format(average_waste))
    print("Highest daily waste: {:.2f} kg".format(highest_waste))
    print("Lowest daily waste: {:.2f} kg".format(lowest_waste))

    print("\n===== DAILY WASTE DETAILS =====")

    for i in range(days):
        print(
            "Day {}: {:.2f} kg".format(
                i + 1, waste_records[i]
            )
        )

    if average_waste > 5:
        print("\nAlert: Average food waste is high.")
        print("Recommendation: Review food preparation quantities.")
    else:
```

```
print("\nFood waste is within the monitored limit.")


print("\nMonitoring completed.")
```

Output:

